North Bengal International University

Lab Report: 03

Exp Name : Performing many types of Queries on database.

Subject : Database management system sessional

Subject Code : CSE 0612-3122

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Rename:

Schema:

SELECT name as TN, Course_id CID
FROM instructor, teacher
WHERE instructor.ID = teacher.ID

Showing rows 0 - 14 (15 total, Query took 0.0005 seconds.)

```
-- ------------------------- -- 8 Test query -- ------------------------- SELECT name AS instructor_name, course_id FROM instructor JOIN teaches ON instructor.ID =
teaches.ID;
```

[Edit inline] [Edit] [Create PHP code]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

String:

Schema:

SELECT dept_name, building
FROM department
WHERE building LIKE '% watson %'

Showing rows 0 - 1 (2 total, Query took 0.0004 seconds.)

```
SELECT dept_name FROM department WHERE building LIKE '%Watson%';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Ordering:

Schema:

SELECT name, ID
FROM instructor
WHERE dept_name = 'physics'
ORDER BY name;

Showing rows 0 - 1 (2 total, Query took 0.0004 seconds.) [name: **EINSTEIN... - GOLD...**]

```
SELECT name FROM instructor WHERE `dept_name` = 'Physics' ORDER BY name;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Union:

Schema:

(SELECT course_id, building, time_slot_id
FROM Section
WHERE semester = 'Fall' AND year = 2009)
UNION
(SELECT course_id, building, time_slot_id
FROM Section
WHERE semester = 'Spring' AND year = 2010)

Showing rows 0 - 7 (8 total, Query took 0.0006 seconds.)

```
(SELECT course_id FROM section WHERE semester = 'Fall' AND year = 2009) UNION (SELECT course_id FROM section WHERE semester = 'Spring' AND year = 2010);
```

[Edit inline] [Edit] [Create PHP code]

Intersect:

Schema:

(SELECT course_id
FROM Section
WHERE semester = ‘Fall’ AND year = 2009)
INTERSECT
(SELECT course_id
FROM Section
WHERE semester = ‘Spring’ AND year = 2010)

Showing rows 0 - 0 (1 total, Query took 0.0005 seconds.)

```
SELECT course_id FROM section WHERE semester = 'Fall' AND year = 2009 AND course_id IN ( SELECT course_id FROM section WHERE semester = 'Spring' AND year = 2010 );
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Except:

Schema:

(SELECT course_id
FROM Section
WHERE semester = ‘Fall’ AND year = 2009)
EXCEPT
(SELECT course_id
FROM Section
WHERE semester = ‘Spring’ AND year = 2010)

Showing rows 0 - 1 (2 total, Query took 0.0006 seconds.)

```
SELECT course_id FROM section WHERE semester = 'Fall' AND year = 2009 EXCEPT SELECT course_id FROM section WHERE semester = 'Spring' AND year = 2010;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Conclusion: In this lab, we practiced several relational algebra and SQL operations, including rename, string functions, ordering, and set-based operations such as union, intersect, and except. These queries helped us understand how to manipulate, filter, and combine data effectively within a database system. Overall, the lab improved our ability to apply fundamental query operations for efficient data retrieval and analysis.

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